

WILLIAM OVERMAN

655 Knight Way, Stanford, CA 94305, USA

✉ wpo@stanford.edu  woverman.com  Google Scholar  LinkedIn

💡 RESEARCH INTERESTS

AI safety and alignment; reinforcement learning; causal inference; healthcare operations.

🎓 EDUCATION

Stanford University, Graduate School of Business 2022–2027 (*expected*)
Ph.D., Operations, Information and Technology
Dissertation defended May 2026; degree conferral pending.

University of California, Irvine 2020–2022
M.Sc., Computer Science

California Institute of Technology (Caltech) 2016–2020
B.S., Mathematics & Computer Science (Double Major)

🏢 INDUSTRY AND RESEARCH POSITIONS

Uber

PhD Scientist Intern, Uber Eats 2025, New York, NY
PhD Scientist Intern, Rider Science 2024, San Francisco, CA
PhD Scientist Intern, Airports 2023, Sunnyvale, CA

Institute for Basic Science

Visiting Researcher 2021, Daejeon, South Korea
Summer Undergraduate Research Fellowship 2019, Daejeon, South Korea

Oxford University, Mathematical Institute

Summer Undergraduate Research Fellowship 2018, Oxford, United Kingdom

📄 PUBLICATIONS

[JOURNAL PAPERS AND SUBMISSIONS]

- **The Oversight Game: Learning to Cooperatively Balance an AI Agent’s Safety and Autonomy**
W. Overman, M. Bayati.
Submitted to *Operations Research*.
Conference version: ICML 2026.
- **Causal Effects with Unobserved Unit Types in Interacting Human–AI Systems**
W. Overman, S. Shirani, M. Bayati.
In preparation for submission to *Manufacturing & Service Operations Management*.
Conference version: NeurIPS 2026.
- **Occupancy Prediction with Patient Data: Evaluating Time-Series, Patient-Level Aggregation, and Deep Set Models**
S.-H. Kim, W. Overman, J. Pauphilet, W. Cha. (alphabetical, sole student).
Major revision at *Manufacturing & Service Operations Management*.
- **Can We Validate Counterfactual Estimations in the Presence of General Network Interference?**
S. Shirani, Y. Luo, W. Overman, R. Xiong, M. Bayati.
Major revision at *Management Science*.
Oral presentation, *Conference on Digital Experimentation @ MIT (CODE@MIT)*, 2025.
- **Some Ordered Ramsey Numbers of Graphs on Four Vertices**
W. Overman, J. Alm, K. Coffey, C. Langhoff.
Australasian Journal of Combinatorics, 2024.

[REFEREED CONFERENCE PAPERS]

- **Causal Effects with Unobserved Unit Types in Interacting Human–AI Systems**
W. Overman, S. Shirani, M. Bayati.
NeurIPS 2026: Neural Information Processing Systems.
Early version: ICML 2026 Workshop on Technical AI Governance Research.
- **The Oversight Game: Learning to Cooperatively Balance an AI Agent’s Safety and Autonomy**
W. Overman, M. Bayati.
ICML 2026: International Conference on Machine Learning.
Early version: NeurIPS 2025 ML×OR Workshop (Oral).
- **Calibrating Conservatism for Scalable Oversight**
W. Overman, M. Bayati.
ICML 2026: International Conference on Machine Learning.
- **Annealed Softmax Greedy in Many-Armed Bayesian Bandits**
W. Overman, M. Bayati.
RLC 2026: Reinforcement Learning Conference.
- **Conformal Arbitrage: Risk-Controlled Balancing of Competing Objectives in Language Models**
W. Overman, M. Bayati.
NeurIPS 2025: Neural Information Processing Systems.
- **Improved Regret Bound for Safe Reinforcement Learning via Tighter Cost Pessimism and Reward Optimism**
K. Yu, D. Lee, W. Overman, D. Lee.
RLC 2025: Reinforcement Learning Conference.
- **Aligning Model Properties via Conformal Risk Control**
W. Overman, J. Vallon, M. Bayati.
NeurIPS 2024: Neural Information Processing Systems.
- **Higher-Order Causal Message Passing for Experimentation with Complex Interference**
M. Bayati, Y. Luo, W. Overman, S. Shirani, R. Xiong. (alphabetical).
NeurIPS 2024: Neural Information Processing Systems.
- **Beating Price of Anarchy and Gradient Descent without Regret in Potential Games**
I. Sakos, S. Leonardos, S. Stavroulakis, W. Overman, I. Panageas, G. Piliouras.
ICLR 2024: International Conference on Learning Representations.
- **Global Convergence of Multi-Agent Policy Gradient in Markov Potential Games**
S. Leonardos, W. Overman, I. Panageas, G. Piliouras (alphabetical, sole student).
ICLR 2022: International Conference on Learning Representations.
- **Independent Natural Policy Gradient Always Converges in Markov Potential Games**
R. Fox, S. McAleer, W. Overman, I. Panageas (alphabetical, sole student).
AISTATS 2022: International Conference on Artificial Intelligence and Statistics.

[WORKING PAPERS]

- **Planning Against Learning in Rank-1 Games**
W. Overman.
- **On Aligning Prediction Models with Clinical Experiential Learning: A Prostate Cancer Case Study**
J. J. Vallon, W. Overman, W. Xu, N. Panjwani, X. Ling, S. Vij, H. P. Bagshaw, J. T. Leppert, S. Shah, G. Sonn, S. Srinivas, E. Pollom, M. K. Buyyounouski, M. Bayati.

☆ AWARDS & SERVICE

Finalist, INFORMS Pierskalla Best Paper Competition	2026
Silver Reviewer Award, ICML	2026
Thomas A. Tisch Prize for Undergraduate Teaching in Computing and Mathematical Sciences Caltech	2020
Dean's Award, UC Irvine, for outstanding research potential	2020
Morgan Ward Prize, Caltech, best original work in mathematics by an underclassman	2018
Mentor, Supervised Program for Alignment Research (SPAR)	2026
Mentoring a six-person team of students and software engineers transitioning into AI alignment research.	
Conference Reviewer: NeurIPS (2024, 2025), ICLR (2026), AISTATS (2026).	

✍ TEACHING

Teaching assistant for all courses listed below.

Stanford University, Graduate School of Business

Optimization and Simulation Modeling	Fall 2026, Fall 2023
Business Intelligence from Big Data	Winter 2026
Operations	Fall 2024

University of California, Irvine

Introduction to Algorithms	Spring 2022
Introduction to Cryptography	Winter 2021

California Institute of Technology (Caltech)

Introduction to Algorithms	Spring 2020, Spring 2019, Spring 2018
Decidability and Tractability	Winter 2020
Discrete Mathematics	Fall 2019, Fall 2018

🎤 INVITED PRESENTATIONS

Seoul National University Data Science Colloquium, Seoul, South Korea	Sep. 2026
Control of Increasingly Capable AI Systems with Safety Guarantees	
NeurIPS ML×OR Workshop Spotlight Presentation, San Diego, CA	Dec. 2025
The Oversight Game: Learning AI Control & Corrigibility in Markov Games	
KAIST ISE Winter Symposium, Daejeon, South Korea	Dec. 2025
Control and Oversight of Black-Box AI Models with Formal Guarantees	
Seoul National University Data Science Seminar, Seoul, South Korea	Oct. 2025
Black-Box Model Alignment: Why It Matters and How We Can Approach It	
KAIST AI Invited Talk, Seoul, South Korea	Sep. 2025
Alignment of Black-Box Models via Conformal Risk Control	
Seoul National University GSDS Invited Talk, Seoul, South Korea	Sep. 2025
Alignment of Black-Box Models via Conformal Risk Control	
GSB Bridging Humans and Machines: Advancing Alignment in AI, Stanford, CA	Apr. 2025
Aligning Model Properties via Conformal Risk Control	
INFORMS Annual Meeting, Seattle, WA	Oct. 2024
Aligning Model Properties via Conformal Risk Control	
Science Jam, Uber Technologies, San Francisco, CA	Aug. 2024
Testing for Network Interference in A/B Tests	
INFORMS Annual Meeting, Phoenix, AZ	Oct. 2023
Calibrate to Operate: Aligning Patient-Level Predictions with System-Wide Forecasts	
Discrete Math Seminar, IBS Discrete Mathematics Group, Daejeon, South Korea	Apr. 2021
Some Ordered Ramsey Numbers of Graphs on Four Vertices	
Caltech CS Theory Seminar, Pasadena, CA	May 2020
Boolean Functions on the Infinite Hamming Cube	